British Applied College
الكلية البريطانية التطبيقية

Guidelines for Accessibility and Quality of IT Systems

Ensuring Inclusive and High-Quality Digital Experiences

Mariam Sameh
The British Applied College

British Applied College
الكلية البريطانية التطبيقية

What We Will Cover Today

- **Introduction** to IT Standards
- **ISO Standards** for IT Quality & Accessibility
- **Deep Dive:** WCAG 1.0 vs. WCAG 2.0
- **The POUR Principles** of WCAG 2.0
- **Interactive Activities** & Quiz
- **Case Study** & Group Discussion

British Applied College
الكلية البريطانية التطبيقية

Learning Objectives

- **Understand** the role of ISO in IT system quality.
- **Identify** key ISO standards (25010, 29110, 24773, 27001).
- **Compare** WCAG 1.0 and WCAG 2.0 frameworks.
- **Apply** the POUR principles to real-world scenarios.

British Applied College
الكلية البريطانية التطبيقية

SECTION 1

ISO Standards for IT Systems

Foundations of Quality, Safety, and Efficiency

ISO: Setting Global Benchmarks

- **Non-governmental organization** that develops and publishes international standards worldwide.
- **Primary Goal:** To ensure the quality, safety, and efficiency of products and services across various industries.
- **In IT Systems:** ISO provides essential guidelines for the development, implementation, and maintenance of accessible technology.

British Applied College
الكلية البريطانية التطبيقية

Essential ISO Frameworks for IT

ISO/IEC 25010:2011

SQuaRE: A comprehensive framework for measuring and evaluating the **quality** of software products and systems.

ISO/IEC 29110

Tailored guidelines for **Very Small Entities (VSEs)**, ensuring accessibility and quality in smaller development environments.

ISO/IEC 24773:2017

Specific guidelines for ensuring the **accessibility** of IT systems for people with disabilities.

ISO/IEC 27001:2013

Requirements for **Information Security Management Systems (ISMS)**, ensuring the quality and security of IT infrastructure.

British Applied College
الكلية البريطانية التطبيقية

Activity: Think-Pair-Share

Think

Why would a small startup need ISO/IEC 29110? Consider quality vs. cost.

Pair

Discuss your thoughts with your neighbor for 2 minutes.

Share

Be ready to share one key benefit with the entire class.

"Standards are not just for giants; they are the foundation for growth."

British Applied College
الكلية البريطانية التطبيقية

SECTION 2

Web Content Accessibility Guidelines (WCAG)

Making the Web Inclusive for Everyone

WCAG 1.0: The First Step (1999)

British Applied College
الكلية البريطانية التطبيقية

Published in **1999** by the World Wide Web Consortium (W3C) as the first set of accessibility guidelines.

Organized around **three priority levels** to define the degree of accessibility compliance.

Priority 1: Essential

Priority 2: Important

Priority 3: Optional

Primarily focused on **HTML-specific technologies**, which limited its application as the web evolved.

Why did we need an update?

British Applied College
الكلية البريطانية التطبيقية

- **Technology-Specific:** Guidelines were heavily focused on specific technologies like **HTML**, making them less adaptable.

- **Limited Scope:** It did not adequately cover modern web technologies such as **CSS and JavaScript**.

- **Complexity:** The guidelines were often difficult to test, verify, and implement consistently across different platforms.

WCAG 2.0: The Gold Standard

- **Published in 2008:** A more comprehensive and up-to-date set of guidelines than its predecessor, WCAG 1.0.
- **Technology-Neutral:** Applies to a wide range of web technologies, including HTML, CSS, and JavaScript.
- **POUR Principles:** Organized around four core principles: Perceivable, Operable, Understandable, and Robust.
- **Success Criteria:** Includes measurable criteria that define what is needed to conform to each guideline.

British Applied College
الكلية البريطانية التطبيقية

The POUR Principles

P

Perceivable

Information and user interface components must be presentable to users in ways they can **perceive** (see or hear).

O

Operable

User interface components and navigation must be **operable**. The interface cannot require interaction that a user cannot perform.

U

Understandable

Information and the operation of the user interface must be **understandable** to all users.

R

Robust

Content must be **robust** enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies.

British Applied College
الكلية البريطانية التطبيقية

Deep Dive: Perceivable & Operable

Perceivable

- **Text Alternatives:** Providing alt-text for non-text content like images.
- **Captions:** Including synchronized captions for all pre-recorded audio/video.
- **Adaptable:** Ensuring content can be presented in different ways without losing info.
- **Distinguishable:** High color contrast and resizable text for easier viewing.

Operable

- **Keyboard Accessible:** All functionality must be available via a keyboard.
- **Enough Time:** Providing users enough time to read and use the content.
- **Seizures:** Avoiding design that is known to cause seizures (e.g., flashing).
- **Navigable:** Providing ways to help users navigate and find content easily.

British Applied College
الكلية البريطانية التطبيقية

Deep Dive: Understandable & Robust

Understandable

- **Readable:** Making text content readable and understandable (e.g., simple language).
- **Predictable:** Making Web pages appear and operate in predictable ways.
- **Input Assistance:** Helping users avoid and correct mistakes (e.g., clear error messages).
- **Consistency:** Using consistent navigation and identification across the site.

Robust

- **Compatible:** Maximize compatibility with current and future user agents.
- **Assistive Tech:** Ensuring content works reliably with screen readers and other tools.
- **Valid Code:** Using valid HTML/CSS to ensure browsers can interpret content correctly.
- **Future-Proof:** Designing content that remains accessible as technologies evolve.

Activity: Fix the Website!

British Applied College
الكلية البريطانية التطبيقية

The Scenario

"A university portal has a video lecture but no captions. Additionally, students must use a mouse to click a small 'Submit' button on an exam page."

Identify: Which POUR principles are being violated here?

Solve: How would you fix these issues to ensure full accessibility?

Discuss: Share your solutions with the class.

Case Study: The Cost of Inaccessibility

British Applied College
الكلية البريطانية التطبيقية

The Example

A major international airline faced a high-profile lawsuit because their booking website was not compatible with screen-reading software.

The Outcome

The company was required to pay significant legal fines, invest in immediate technical overhauls, and suffered damage to their brand reputation.

The Lesson

Accessibility is not just a "nice to have" feature; it is a fundamental legal, ethical, and professional requirement for all digital platforms.

> ***"Inaccessibility is a barrier to opportunity and a risk to business."***

Quick Poll: Accessibility Awareness

British Applied College
الكلية البريطانية التطبيقية

Have you ever used "Dark Mode" on your devices?

- [ ] Yes, always
- [ ] Sometimes
- [ ] Never

Have you ever watched a video with captions in a loud place?

- [ ] Yes, frequently
- [ ] Occasionally
- [ ] No

*Accessibility benefits **everyone**, not just people with permanent disabilities.*

Knowledge Check: ISO Standards

British Applied College
الكلية البريطانية التطبيقية

Which ISO standard focuses on Information Security Management Systems (ISMS)?

A) ISO/IEC 25010

B) ISO/IEC 29110

C) ISO/IEC 27001

D) ISO/IEC 24773

Think about the standard that ensures both quality and security.

Knowledge Check: POUR Principles

British Applied College
الكلية البريطانية التطبيقية

Providing "Alternative Text" for an image falls under which accessibility principle?

A) Perceivable

B) Operable

C) Understandable

D) Robust

Remember: This principle ensures users can see or hear the content.

Reflection: Designing for All

"How can we, as future IT professionals, ensure accessibility is part of the start of a project, not an afterthought?"

Brainstorming Session (3 Minutes): Turn to your group and list three practical ways to integrate accessibility into the initial design phase.

British Applied College
الكلية البريطانية التطبيقية

Key Takeaways

ISO Standards

Provide the essential global framework for ensuring IT system quality, security, and accessibility.

WCAG Evolution

WCAG 1.0 started the movement, while WCAG 2.0 perfected the principles for modern web technologies.

POUR Principles

Perceivable, Operable, Understandable, and Robust serve as your definitive checklist for accessibility.

A Professional Necessity

British Applied College
الكلية البريطانية التطبيقية

Thank You!

Any Questions?

Mariam Sameh

The British Applied College